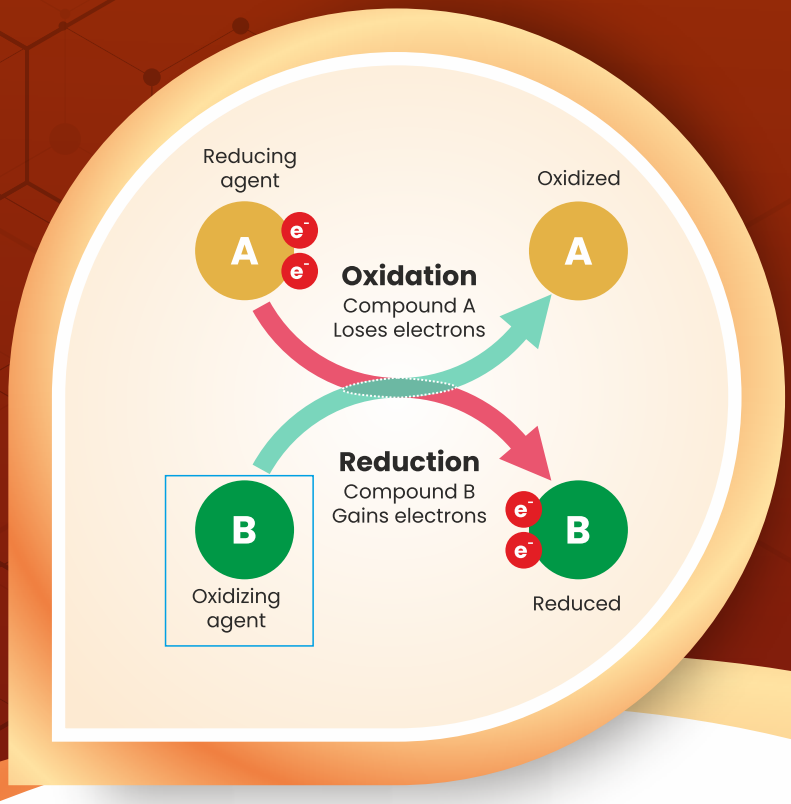
ALLEN CAREER INSTITUTE
KOTA (RAJASTHAN)
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PHYSICAL CHEMISTRY

JEE (MAIN+ADVANCED)

NURTURE COURSE



Exercise

Redox & Equivalent Concepts

(English Medium)

EXERCISE # S-I

OXIDATION NUMBER

1. Calculate oxidation number of the underlined elements in the following compounds or ions

(a) $\underline{H_4}P_2O_7$ (b) $\underline{S_2}O_3^{2-}$ (c) $Ca\underline{O_2}$ (d) $\underline{Zn}O_2^{2-}$

(e) $\underline{Cr}O_2Cl_2$ (f) $Ca(\underline{Cl}O_2)_2$ (g) $\underline{C_3}O_2$ (h) $K\underline{I_3}$

(i) $\underline{Fe_3}O_4$ (j) $\underline{Br_3}O_8$ (k) $CaO\underline{Cl_2}$ (l) $\underline{O_2}^+$

(m) $\underline{O_2}^-$ (n) $\underline{C}N^-$

RR0179

2. Calculate oxidation number of the underlined elements in the following compounds or ions

(a) $K[\underline{Co}(C_2O_4)_2.(NH_3)_2]$ (b) $K\underline{Al}(\underline{S}O_4)_2.12H_2O$

(c) $[\underline{Fe}(CN)_6]^{4-}$ (d) $\underline{Fe}_{0.93}O$

(e) $H_2\underline{S}_2O_8$ (f) $\underline{Cr}O_5$

(g) $H_2\underline{S}O_5$

RR0180

OXIDATION - REDUCTION

3. Classify each of the following unbalanced half–reactions as either an oxidation or a reduction :

RR0181

(a) $O_2(g) \longrightarrow OH^-(aq)$ (b) $H_2O_2(aq) \longrightarrow O_2(g)$

(c) $MnO_4^-(aq) \longrightarrow MnO_4^{2-}(aq)$ (d) $CH_3OH(aq) \longrightarrow CH_2O(aq)$

(e) $MnO_2(s) \longrightarrow Mn_3O_4(s)$ (f) $FeO(s) \longrightarrow Fe_3O_4(s)$

(g) $CH_3COOH\ (aq.) \longrightarrow CH_3CH_2OH(aq)$

OXIDISING AND REDUCING AGENT

4. Identify the oxidant (oxidising agent) and reductant (reducing agent) in the following reacions

RR0182

(a) $Zn(s) + \frac{1}{2}O_2(g) \longrightarrow ZnO(s)$

(b) $Zn(s) + 2H^+(aq) \longrightarrow Zn^{2+}(aq) + H_2(g)$

(c) $10H^+(aq) + 4Zn(s) + NO_3^-(aq) \longrightarrow 4Zn^{2+}(aq) + NH_4^+(aq) + 3H_2O(\ell)$

(d) $2H_2S\ (aq) + SO_2(g) \rightarrow 3S(s) + 2H_2O\ (l)$

(e) $2MnO_4^-\ (aq) + 16H^+\ (aq) + 10\ Cl^-(aq.) \longrightarrow 2Mn^{2+}(aq.) + 8H_2O(l) + 5Cl_2(g)$

(f) $2MnO_4^-\ (aq) + 6H^+\ (aq) + 5H_2O_2(aq.) \longrightarrow 2Mn^{2+}(aq.) + 8H_2O(l) + 5O_2(g)$

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BALANCING OF REDOX

5. Balance the following redox reactions by oxidation number method : **RR0183**

(i) $Cu + HNO_3 \longrightarrow Cu(NO_3)_2 + NO_2 + H_2O$

(ii) $Br_2 + H_2O_2 \xrightarrow{H^+} BrO_3^- + H_2O$

(iii) $MnO_4^- + Fe^{+2} \longrightarrow Mn^{+2} + Fe^{+3}$

(iv) $MnO_2 + Cl^- \xrightarrow{H^+} Mn^{+2} + Cl_2 + H_2O$

(v) $ClO^- + CrO_2^- \xrightarrow{OH^-} Cl^- + CrO_4^{2-} + H_2O$

(vi) $Zn + NaOH \longrightarrow Na_2ZnO_2 + H_2$

(vii) $Na_2S_2O_3 + I_2 \longrightarrow Na_2S_4O_6 + NaI$

6. Balance the following redox reactions by half-reaction method (Ion electron method): **RR0184**

(i) $MnO_4^-(aq) + I^-(aq) \longrightarrow MnO_2(s) + I_2(s)$ (basic medium)

(ii) $MnO_4^-(aq) + SO_2(g) \longrightarrow Mn^{2+}(aq) + HSO_4^-(aq)$ (acidic medium)

(iii) $H_2O_2(aq) + Fe^{2+}(aq) \longrightarrow Fe^{3+}(aq) + H_2O(l)$ (acidic medium)

(iv) $Cr_2O_7^{2-}(aq.) + SO_2(g) \longrightarrow Cr^{3+}(aq) + SO_4^{2-}(aq)$ (acidic medium)

(v) $P_4(s) \longrightarrow PH_3(g) + H_2PO_2^-(aq.)$ (basic medium)

(vi) $MnO_4^- + FeC_2O_4 \xrightarrow{H^+} Mn^{2+} + CO_2 + Fe^{3+}$

(vii) $S_2O_3^{2-} + I_2 \longrightarrow S_4O_6^{2-} + I^-$

n-FACTOR & EQUIVALENT WEIGHT

7. Find the n-factor of underlined species in the following non redox reaction.

(1) $NaOH + \underline{H_3PO_4} \longrightarrow NaH_2PO_4 + H_2O$

(2) $NaOH + \underline{H_2SO_4} \longrightarrow NaHSO_4 + H_2O$

(3) $\underline{Ca(OH)_2} + HCl \longrightarrow Ca(OH)Cl + H_2O$

(4) $\underline{Na_2CO_3} + HCl \longrightarrow NaHCO_3 + NaCl$

(5) $\underline{Na_2CO_3} + HCl \longrightarrow NaCl + H_2O + CO_2$ **RR0014**

8. What will be the equivalent weight of H_3PO_4 in each of the reaction.

(i) $H_3PO_4 + OH^- \longrightarrow H_2PO_4^- + H_2O$

(ii) $H_3PO_4 + 2OH^- \longrightarrow HPO_4^{2-} + 2H_2O$

(iii) $H_3PO_4 + 3OH^- \longrightarrow PO_4^{3-} + 3H_2O$ **RR0015**

9. What is the equivalent weight of HNO_3 in following reaction ?

$HNO_3 + H_2S \longrightarrow H_2O + NO + S$ **RR0016**

10. Find out the equivalent weight of the underlined species in the following reaction :

(i) $\underline{ClO_3^-} + Fe^{2+} + H^+ \longrightarrow Cl^- + Fe^{3+} + H_2O$

(ii) $CuO + \underline{NH_3} \longrightarrow Cu + N_2 + H_2O$ **RR0017**

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11. Find out the n_{factor} of IO_3^- in the following disproportionation reaction.

$$I_2 \xrightarrow{OH^-} IO_3^- + I^-$$ **RR0019**

12. What is the equivalent weight of the following salts ?

(i) Na_2SO_4 (ii) $Na_2CO_3 . 10H_2O$ (iii) $AlCl_3$ (iv) $Al_2(SO_4)_3$

(v) $Fe_2(SO_4)_3.9H_2O$ **RR0185**

13. What are the n-factor of underlined species in the following proceses ?

(i) $\underline{Fe_{0.93}O} \longrightarrow Fe_2O_3$ (ii) $Fe_{0.93}O \longrightarrow \underline{Fe_2O_3}$

(iii) $\underline{Fe_2O_3} \longrightarrow Fe_3O_4$ (iv) $Fe_2O_3 \longrightarrow \underline{Fe_3O_4}$ **RR0186**

14. What are the n-factors of underlines species in the following process ?

(i) $\underline{FeC_2O_4} \longrightarrow Fe_2O_3 + CO_2$ (ii) $\underline{FeS_2} \longrightarrow Fe_2O_3 + SO_2$

(iii) $\underline{CrBr_3} \longrightarrow CrO_4^{2-} + BrO_4^-$ (iv) $\underline{Co(MnO_4)_3} \longrightarrow Co^{2+} + Mn^{2+}$ **RR0187**

15. What are the equivalent weights of H_2SO_4 in the following processes/reactions ?

(i) $Mg + H_2SO_4 \longrightarrow MgSO_4 + SO_2 + H_2O$

(ii) $NaOH + H_2SO_4 \longrightarrow NaHSO_4 + H_2O$

(iii) $H_2SO_4 \longrightarrow H_2S_2O_8$

(iv) $Pb + PbO_2 + H_2SO_4 \longrightarrow PbSO_4 + H_2O$ **RR0188**

ACID BASE TITRATION

16. How many millilitre of 0.5 M H_2SO_4 is needed to dissolve 4.94 g of copper II carbonate ? [Cu = 63.5] **[JEE 1999]**

RR0189

17. An aqueous solution of 6.3 g of oxalic acid dihydrate ($H_2C_2O_4.2H_2O$) is made upto 250 mL. The volume (in ml) of 0.1 N NaOH required to completely neutralise 10 mL of this solution is :

[JEE 2001]

RR0021

18. Calculate volume (in ml) of 1N H_3PO_4 required to react with 20 ml 2N $Ca(OH)_2$ solution ?

RR0022

19. Calculate volume (in L) of 0.4 M NaOH required to react with following mixture

HCl(1 mol) + H_2SO_4 (2 mol) **RR0024**

20. How many litre of 0.1N HCl are required to react completely with 19 gm mixture of Na_2CO_3 and $NaHCO_3$ containing equimolar amount of these two ? **RR0026**

21. H_3PO_4 is a tri basic acid and one of its salt is NaH_2PO_4. What volume (in ml) of 1 M NaOH solution should be added to 12 g of NaH_2PO_4 to convert it into Na_3PO_4 ? **RR0027**

22. A sample containing 4.2g of KOH and $Ca(OH)_2$ is neutralized by an acid. It consumes 0.1 equivalent of acid. Calculate the mass percentage of KOH in the sample. (K = 39) **R00061**

23. What volume (in ml) 0.4 M - HCl solution needed for complete reaction with 3.4 gm Al_2O_3 ?

RR0190

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24. 20 ml of 2M - NaOH solution is diluted to 400 ml. 25 ml of the diluted solution exactly requires 50 ml of HNO_3 solution for titration. What is the molarity of HNO_3 solution ? **RR0191**

25. 85.8 gm washing soda, $Na_2CO_3.10\ H_2O$, is dissolved in sufficient water to get 600 ml solution. What volume (in ml) of this solution will neutralise completely, 40 ml of 0.2 M - H_2SO_4 solution ?

RR0192

REDOX TITRATION

26. A 5.0 cm^3 solution of H_2O_2 liberates 0.508 g of iodine from an acidified KI solution. Calculate the strength of H_2O_2 solution in terms of volume strength at STP. (I = 127) **[JEE' 1995]**

RR0029

27. Calculate the normality of a solution containing 15.8 g of $KMnO_4$ in 50 mL acidic solution.

(K = 39, Mn = 55) **RR0030**

28. 20 mL of H_2O_2 after acidification with dil H_2SO_4 required 30 mL of $\frac{N}{12}$ $KMnO_4$ for complete oxidation. The strength of H_2O_2 solution (in gm/litre) is [Molar mass of H_2O_2 = 34] **RR0031**

29. A volume of 10.0 ml of 1 M SeO_2 reacted with exactly 20 ml of 2 M $CrSO_4$. In the reaction, Cr^{2+} was oxidized to Cr^{3+}. To what oxidation state was selenium converted by the reaction ?

RR0033

30. Potassium acid oxalate $K_2C_2O_4.\ 3H_2C_2O_4.4H_2O$ can be oxidized by MnO_4^- in acid medium. Calculate the volume (in ml) of 0.1 M $KMnO_4$ reacting in acid solution with 5.08 gm of the acid oxalate. (K = 39) **RR0034**

31. A 1 g sample of H_2O_2 solution containing x% H_2O_2 by mass requires x cm^3 of a $KMnO_4$ solution for complete oxidation under acidic condition. Calculate the normality of $KMnO_4$ solution.

RR0035

32. Metallic tin in the presence of HCl is oxidized by $K_2Cr_2O_7$ to stannic chloride, $SnCl_4$. What volume (in litre) of deci-normal dichromate solution would be reduced by 11.9 gm of tin [Sn = 119]

RR0036

33. 520 gm mixture of Fe_2O_3 and FeO reacts completely with 158 gm $KMnO_4$ in acidic medium Calculate the mole % of Fe_2O_3 in mixture. [Fe = 56, K = 39, Mn = 55] **RR0065**

34. Calculate the millimoles of Br_2 produced when 10 ml of 0.1 M BrO_3^- reacts with excess of Br^-.

RR0066

35. What volume (in ml) of 0.5 M-$Na_2S_2O_3$ solution will react completely with 12.7 gm iodine ?

(I = 127) **RR0193**

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BACK TITRATION

36. 10 g $CaCO_3$ were dissolved in 250 ml of 1 M HCl. What volume (in ml) of 2 M KOH would be required to neutralise excess HCl ? **RR0041**

37. 20 ml of 2M -H_2SO_4 solution is mixed with 30 ml of 3M - NaOH solution. Calculate the volume (in ml) of 4M-HCl solution needed to neutralise the resulting solution. **RR0194**

38. 50 ml of 0.02 M -acidified $KMnO_4$ solution is mixed with 40 ml of KI solution. If the resulting solution requires 20 ml of 0.1 M-$K_2Cr_2O_7$ solution for titration of excess KI, calculate the molarity of KI solution. **RR0195**

39. 10 ml of 0.8 M - HNO_3 solution is mixed with 20 ml of 0.2 M - NaOH solution and the mixture is diluted to 100 ml. If 25 ml of the diluted solution exactly require 8 ml of KOH solution for titration, what is the concentration (in gm/litre) of KOH solution used ? (K = 39) **RR0196**

40. 6.3 gm $H_2C_2O_4.2H_2O$ is dissolved in sufficient water to get 500 ml solution. To 50 ml of this solution, 25 ml of (N/5) $KMnO_4$ solution is added in presence H_2SO_4. Calculate the volume (in ml) of (N/6) $K_2Cr_2O_7$ solution needed for titration of excess oxalate. **RR0197**

HARDNESS OF WATER

41. One litre of a sample of hard water contains 11.1 mg of $CaCl_2$ & 9.5 mg of $MgCl_2$. What is degree of hardness in terms of ppm of $CaCO_3$ **RR0045**

42. A sample of hard water contains 324 ppm of $Ca(HCO_3)_2$. How many gm of CaO is needed to remove all bicarbonates from 100 L hard water ? **RR0198**

43. The number of hardness creating salts among NaCl, K_2CO_3, $Mg(HCO_3)_2$, $Mg(NO_3)_2$, Na_3PO_4, $Ca(HCO_3)_2$, $MgBr_2$, is **RR0199**

44. 50 L of a water sample contains 7.3 gm $Mg(HCO_3)_2$. Calculate temporary hardness of the water in terms of ppm of $CaCO_3$. **RR0200**

45. A sample of tap water contains 300 ppm $MgSO_4$, 648 ppm $Ca(HCO_3)_2$ and 585 ppm NaCl. Calculate permanent hardness of tap water in terms of ppm of $CaCO_3$. **RR0201**

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EXERCISE # S-II

1. Find out the oxidation number of the underlined elements in the following compounds:

(a) $\underline{C}H_3\underline{C}H_2OH$ (b) $\underline{C}H_3\underline{C}OOH$ (c) $H\underline{O}F$ (d) $C_6H_5\underline{C}HO$

(e) $C_6H_5\underline{N}O_2$ (f) $C_6H_5\underline{N}H_2$ (g) $(\underline{C}N)_2$ (h) $H\underline{C}N$

(i) $HN\underline{C}$ (j) $\underline{C}OCl_2$ **RR0202**

2. Balance the following redox reactions by half-reaction method

(i) $KMnO_4(aq.) + FeC_2O_4(aq.) + H_2SO_4(aq.) \longrightarrow$

$MnSO_4(aq.) + Fe_2(SO_4)_3(aq.) + CO_2(g) + H_2O(l)$ (Acidic medium) **RR0054**

(ii) $HNO_2(aq.) \rightarrow NO_3^- + NO(g)$ (Acidic medium) **RR0057**

(iii) $IO_3^- + I^- + H^+ \rightarrow I_2 + H_2O$ (Acidic medium) **RR0058**

(iv) $Cu_2S + MnO_4^- + H^+ \rightarrow Cu^{2+} + Mn^{2+} + SO_2$ (Acidic medium) **RR0059**

(v) $HCHO(l) + [Ag(NH_3)_2]^+(aq.) + OH^-(aq.) \rightarrow Ag(s) + HCOO^-(aq.) + NH_3(aq.) + H_2O(l)$
(Basic medium) **RR0060**

3. Find out the n_{factor} of $(NH_4)_2Cr_2O_7$ in the following decomposition reaction.

$(NH_4)_2Cr_2O_7 \longrightarrow Cr_2O_3 + N_2(g) + H_2O$ **RR0018**

4. A mixture of H_2SO_4 and $H_2C_2O_4$ (oxalic acid) and some inert impurity weighing 3.185 g was dissolved in water and the solution made up to 1 litre, 10 mL of this solution required 3 mL of 0.1 N NaOH for complete neutralization. In another experiment 100 mL of the same solution in hot condition required 4 mL of 0.02M $KMnO_4$ solution for complete reaction. The wt. % of H_2SO_4 in the mixture is :- **RR0072**

5. 25 ml of 0.4 M -acidified $KMnO_4$ solution is treated with excess of KI solution. The liberated I_2 exactly requires 20 ml of $Na_2S_2O_3$ solution for titration. What is the molarity of $Na_2S_2O_3$ solution ? **RR0203**

6. 25 ml of 0.4 M -acidified KIO_3 solution is treated with excess of KI solution. The liberated I_2 exactly requires 20 ml of $Na_2S_2O_3$ solution for titration. What is the molarity of $Na_2S_2O_3$ solution ? **RR0204**

7. 17.4 gm pyrolusite ore is dissolved in excess of HCl solution. The liberated Cl_2 gas is passed through excess of KI solution. If the liberated I_2 exactly requires 40 ml of decimolar hypo solution for titration, calculate the percentage of pure MnO_2 in the pyrolusite ore. (Mn = 55). **RR0205**

8. 4.00 gm NaOH is dissolved in 100 ml HCl solution and the solution is diluted to 500 ml. If 25 ml of the diluted solution exactly requires 10 ml of 0.02 M -Na_2CO_3 solution for titration, calculate the concentration of original HCl solution (in gm/litre). **RR0206**

9. 30 ml of '22.7V' H_2O_2 solution is added in 50 ml 0.3 M - $KMnO_4$ solution in presence of excess H_2SO_4 and the resulting solution is diluted to 225 ml. 25 ml of the diluted solution exactly require 20 ml of 0.05 M solution of an oxidising agent for titration. Calculate the moles of electron taken by each mole of oxidising agent. **RR0207**

10. 12.7 gm I_2 is dissolved in 200 ml of 1.5 M - NaOH solution. Calculate the volume (in ml) of 0.8 M - HCl solution needed for complete neutralisation of excess NaOH. [I = 127] **RR0208**

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EXERCISE # O-I

SINGLE CORRECT :

OXIDATION NUMBER

1. The oxidation number of phosphorus in $Ba(H_2PO_2)_2$ is :

(A) +3 (B) +2 (C) +1 (D) –1 **RR0074**

2. In the coordination compound, $K_4[Ni(CN)_6]$, the oxidation state of nickel is **[AIEEE-03]**

(A) +1 (B) +2 (C) –1 (D) 0 **RR0076**

3. The oxidation state of Cr in $[Cr(NH_3)_4Cl_2]^+$ is - **[AIEEE-05]**

(A) +2 (B) +3 (C) 0 (D) +1 **RR0077**

4. Oxidation number of Cl in $CaOCl_2$ (bleaching powder is) **[AIEEE-02]**

(A) Zero, since it contains Cl_2 (B) –1, since it contains Cl^-

(C) +1, since it contains ClO^- (D) +1 and –1 since it contains ClO^- and Cl^-

RR0078

5. The oxidation number of sulphur in S_8, S_2F_2 and H_2S respectively are :

(A) 0, +1 and –2 (B) +2, +1 and –2

(C) 0, +1 and +2 (D) –2, +1 and –2 **RR0079**

6. The incorrect order of decreasing oxidation number of S in compounds is :-

(A) $H_2S_2O_7 > Na_2S_4O_6 > Na_2S_2O_3 > S_8$ (B) $H_2SO_5 > H_2SO_3 > SCl_2 > H_2S$

(C) $SO_3 > SO_2 > S_8 > H_2S$ (D) $H_2SO_4 > SO_2 > H_2S > H_2S_2O_8$ **RR0080**

7. Match List-I (Compounds) with List-II (Oxidation states of nitrogen) and select answer using the codes given below the lists :-

List-I	List-II
(a) NaN_3	1. +5
(B) N_2H_2	2. +2
(C) NO	3. –1/3
(D) N_2O_5	4. –1

Code :	(a)	(B)	(C)	(D)
(A)	3	4	2	1
(B)	4	3	2	1
(C)	3	4	1	2
(D)	4	3	1	2

RR0085

OXIDATION REDUCTION

8. Which of the following is a redox reation ? **[AIEEE-02]**

(A) $2NaAg(CN)_2 + Zn \longrightarrow Na_2Zn(CN)_4 + 2Ag$

(B) $BaO_2 + H_2SO_4 \longrightarrow BaSO_4 + H_2O_2$

(C) $N_2O_5 + H_2O \longrightarrow 2HNO_3$

(D) $AgNO_3 + KI \longrightarrow AgI + KNO_3$ **RR0081**

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9. Which reaction does not represent a disproportionation reaction ?

(A) $Cl_2 + OH^- \longrightarrow Cl^- + ClO_3^- + H_2O$ (B) $2H_2O_2 \longrightarrow H_2O + O_2$

(C) $2Cu^+ \longrightarrow Cu^{+2} + Cu$ (D) $(NH_4)_2Cr_2O_7 \longrightarrow N_2 + Cr_2O_3 + 4H_2O$

RR0082

10. Which of the following is not a redox reaction?

(A) $BaO_2 + H_2SO_4 \rightarrow BaSO_4 + H_2O_2$ (B) $2BaO + O_2 \rightarrow 2BaO_2$

(C) $2KClO_3 \rightarrow 2KCl + 3O_2$ (D) $SO_2 + 2H_2S \rightarrow 2H_2O + 3S$ **RR0083**

11. The molecular weight of compounds ?

(a) Na_2SO_4 (b) $Na_3PO_4.12H_2O$ (c) $Ca_3(PO_4)_2$ respectively are X, Y and Z.

The correct set of equivalent weight will be :

(A) $\frac{X}{2}, \frac{Y}{3}, \frac{Z}{6}$ (B) $\frac{X}{2}, Y, \frac{Z}{3}$ (C) $X, \frac{Y}{3}, \frac{Z}{3}$ (D) X, Y, Z

RR0245

12. H_2O_2 acts as a reducing agent in:

(A) $FeCl_2 + HCl + H_2O_2 \longrightarrow FeCl_3 + H_2O$ (B) $Cl_2 + H_2O_2 \longrightarrow HCl + O_2$

(C) $HI + H_2O_2 \longrightarrow I_2 + H_2O$ (D) $H_2SO_3 + H_2O_2 \longrightarrow H_2SO_4 + H_2O$

RR0084

13. MnO_4^- is good oxidising agent in different medium changing to - **[AIEEE-02]**

$MnO_4^- \longrightarrow Mn^{2+}$

$\longrightarrow MnO_4^{2-}$

$\longrightarrow MnO_2$

$\longrightarrow Mn_2O_3$

Changes in oxidation number respectively are -

(A) 1, 3, 4, 5 (B) 5, 4, 3, 2 (C) 5, 1, 3, 4 (D) 2, 6, 4, 3 **RR0086**

14. The oxidation state of chromium in the final product formed by the reaction between Kl and acidified potassium dichromate solution is - **[AIEEE-05]**

(A) +6 (B) +4 (C) +3 (D) +2 **RR0104**

BALANCING OF REDOX REACTIONS

15. During the disproportionation of iodine to iodide and iodate ions, the ratio of iodate and iodide ions formed in alkaline medium is :-

(A) 1 : 5 (B) 5 : 1 (C) 3 : 1 (D) 1 : 3 **RR0124**

16. In the reaction,

$xHI + yHNO_3 \longrightarrow NO + I_2 + H_2O$

(A) x = 3, y = 2 (B) x = 2, y = 3 (C) x = 6, y = 2 (D) x = 6, y = 1 **RR0087**

17. The number of electrons to balance the following equation :-

$NO_3^- + 4H^+ + e^- \rightarrow 2H_2O + NO$ is

(A) 5 (B) 4 (C) 3 (D) 2 **RR0088**

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18. For the redox reaction,

$$MnO_4^- + C_2O_4^{2-} + H^+ \rightarrow Mn^{2+} + CO_2 + H_2O$$

the correct coefficients of the reactants for the balanced reaction are :

	MnO_4^-	$C_2O_4^{2-}$	H^+	
(A)	2	5	16	
(B)	16	5	2	
(C)	5	16	2	
(D)	2	16	5	**RR0090**

19. When arsenic sulphide is boiled with NaOH, sodium arsenite and sodium thioarsenite are formed

$x\ As_2S_3 + y\ NaOH \longrightarrow Na_3AsO_3 + x\ Na_3AsS_3 + \frac{y}{2} H_2O$. What are the values of x and y ?

(A) 1, 6 (B) 2, 8 (C) 2, 6 (D) 1, 4 **RR0125**

20. Number of moles of electrons taken up when 1 mole of NO_3^- ions is reduced to 1 mole of NH_2OH is

(A) 2 (B) 4 (C) 5 (D) 6 **RR0089**

n-FACTOR AND EQUIVALENT WEIGHT

21. When N_2 is converted into NH_3, the equivalent weight of nitrogen will be :

(A) 1.67 (B) 2.67 (C) 2.33 (D) 4.67 **RR0092**

22. The equivalent mass of $MnSO_4$ is half its molecular mass when it is converted to : **[JEE 1998]**

(A) Mn_2O_3 (B) MnO_2 (C) MnO_4^- (D) MnO_4^{2-} **RR0093**

23. When $KBrO_3$ ion reacts with Br^- ion in acid solution Br_2 is liberated.The equivalent weight of $KBrO_3$ in this reaction is

(A) M/8 (B) M/3 (C) M/5 (D) M/6 **RR0094**

24. A solution of $KMnO_4$ is reduced to MnO_2. The normality of $KMnO_4$ solution is 0.6. The molarity is

(A) 1.8M (B) 0.6M (C) 0.1M (D) 0.2M **RR0096**

25. 0.52 g of a dibasic acid required 100 mL of 0.2 N NaOH for complete neutralization. The equivalent weight of acid is

(A) 26 (B) 52 (C) 104 (D) 156 **RR0098**

26. What is the equivalent weight of H_2SO_4 in the reaction ?

$$H_2SO_4 + NaI \rightarrow Na_2SO_4 + I_2 + H_2S + H_2O$$

(A) 12.25 (B) 49 (C) 61.25 (D) 98.00 **RR0099**

27. The mass of oxalic acid crystals ($H_2C_2O_4.2H_2O$) required to prepare 50 mL of a 0.2 N solution is:-

(A) 4.5 g (B) 6.3 g (C) 0.63 g (D) 0.45 g **RR0101**

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ACID - BASE TITRATION

28. If 25 mL of a H_2SO_4 solution reacts completely with 1.06 g of pure Na_2CO_3, what is the normality of this acid solution :

(A) 1 N (B) 0.5 N (C) 1.8 N (D) 0.8 N **RR0102**

29. 100 ml of 0.1M $NaAl(OH)_2CO_3$ is neutralised by 0.25 N HCl to form NaCl, $AlCl_3$ and CO_2. Volume of HCl required is

(A) 10 mL (B) 40 mL (C) 100mL (D) 160 mL **RR0116**

REDOX TITRATION

30. The number of mole of $KMnO_4$ that will need to react completely with one mole ferrous oxalate in acidic solution is : **[JEE 1997]**

(A) 2/5 (B) 3/5 (C) 4/5 (D) 1 **RR0105**

31. The number of mole of $KMnO_4$ that will be needed to react with one mole of sulphite ion in acidic solution is : **[JEE 1997]**

(A) 2/5 (B) 3/5 (C) 4/5 (D) 1 **RR0106**

32. A mixture of $NaHC_2O_4$ and $KHC_2O_4 . H_2C_2O_4$ required equal volumes of 0.2 N $KMnO_4$ and 0.12 N NaOH separately. What is the molar ratio of $NaHC_2O_4$ and $KHC_2O_4 . H_2C_2O_4$ in the mixture ?

(A) 6 : 1 (B) 1 : 2 (C) 3 : 1 (D) 4 : 1 **RR0112**

33. Volume V_1 mL of 0.1 M $K_2Cr_2O_7$ is needed for complete oxidation of 0.678 g N_2H_4 in acidic medium. The volume of 0.3 M $KMnO_4$ needed for same oxidation in acidic medium will be :-

(A) $\frac{2}{5}V_1$ (B) $\frac{5}{2}V_1$ (C) $\frac{1}{3}V_1$ (D) $\frac{5}{6}V_1$ **RR0113**

34. As_2O_3 is oxidised to H_3AsO_4 by $KMnO_4$ in acidic medium. Volume of 0.02M $KMnO_4$ required for this purpose by 1mmol of As_2O_3 will be

(A) 10 mL (B) 20 mL (C) 40 mL (D) 80 mL **RR0114**

35. x mmol of XeF_4 quantitatively oxidized KI to I_2 and liberated Xe, alongwith formation of KF. This iodine required 20 ml of decinormal hypo solution for exact titration. The value of x is

(A) 0.5 (B) 1.0 (C) 2.0 (D) 5.0 **RR0117**

36. 0.3 g of an oxalate salt was dissolved in 100 mL solution. The solution required 90 mL of N/20 $KMnO_4$ for complete oxidation. The % of oxalate ion in salt is :-

(A) 33% (B) 66% (C) 70% (D) 40% **RR0119**

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37. The number of moles of $Cr_2O_7^{2-}$ needed to oxidize 0.136 equivalents of $N_2H_5^+$ by the reaction $N_2H_5^+ + Cr_2O_7^{2-} \rightarrow N_2 + Cr^{3+} + H_2O$ is

(A) 0.136 (B) 0.068 (C) 0.0227 (D) 0.272 **RR0121**

38. HNO_3 oxidises NH_4^+ ions to nitrogen and itself gets reduced to NO_2. The moles of HNO_3 required by 1 mol of $(NH_4)_2SO_4$ is :-

(A) 4 (B) 5 (C) 6 (D) 2 **RR0123**

HARDNESS OF WATER

39. Which of the following salt does not cause hardness in water ?

(A) $Ca(NO_3)_2$ (B) K_2SO_4 (C) $MgSO_4$ (D) $MgCl_2$ **RR0209**

40. Distilled water may be obtained from hard water by -

(A) Calgon process (B) Zeolite process

(C) Clark process (D) Ion-exchange resin process **RR0210**

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EXERCISE # O-II

SINGLE CORRECT :

1. If 10 g of V_2O_5 is dissolved in acid and is reduced to V^{2+} by zinc metal, how many mole I_2 could be reduced by the resulting solution if it is further oxidised to VO^{2+} ions ?

[Assume no change in state of Zn^{2+} ions] (V = 51, O = 16, I = 127) :

(A) 0.11 mole of I_2 (B) 0.22 mole of I_2

(C) 0.055 mole of I_2 (D) 0.44 mole of I_2 **RR0120**

2. Dichloroacetic acid ($CHCl_2COOH$) is oxidized to CO_2 and Cl_2 by 600 meq of an oxidizing agent. Same amount of acid can neutralize how many moles of ammonia to form ammonium dichloroacetate?

(A) 0.0167 (B) 0.1 (C) 0.3 (D) 0.6 **RR0122**

3. An element A in a compound ABD has oxidation number A^{n-}. It is oxidised by $Cr_2O_7^{2-}$ in acid medium. In the experiment 1.68×10^{-3} moles of $K_2Cr_2O_7$ were used for 3.36×10^{-3} moles of ABD. The new oxidation number of A after oxidation is :-

(A) 3 (B) 3 – n (C) n – 3 (D) +n3 **RR0127**

4. A sample of $KMnO_4$ solution required 50 ml when titrated against 3 mmol of oxalic acid. The normality of same solution in reaction with alkaline H_2O_2 is

(A) 0.120 N (B) 0.060 N (C) 0.072 N (D) 0.036 N **RR0211**

5. In the reaction $CrO_5 + H_2SO_4 \rightarrow Cr_2(SO_4)_3 + H_2O + O_2$ one mole of CrO_5 will liberate how many moles of O_2 ?

(A) 5/2 (B) 5/4 (C) 9/2 (D) 7/4 **RR0095**

ONE OR MORE THAN ONE CORRECT :

6. Which of the following process(es) is/are either oxidation or reduction ?

(A) $S_2O_3^{2-} \rightarrow S_4O_6^{2-}$ (B) $C_4H_6 \rightarrow CO_2$

(C) $Cr_2O_7^{2-} \rightarrow CrO_4^{2-}$ (D) $H_2SO_4 \rightarrow H_2S_2O_8$ **RR0212**

7. The correct information(s) about the n-factor of given species in the reaction :

$KMnO_4 + H_2SO_4 + HCl \rightarrow K_2SO_4 + MnSO_4 + Cl_2 + H_2O$ is/are -

(A) $KMnO_4 \rightarrow 5$ (B) $H_2SO_4 \rightarrow 10/3$

(C) $HCl \rightarrow 1$ (D) $K_2SO_4 \rightarrow 2$ **RR0213**

8. 0.2 mole of Cl_2 is reacted with 500 ml of 1.2 M - NaOH solution

$Cl_2 + OH^- \rightarrow ClO_3^- + Cl^- + H_2O$

The incorrect information(s) related is/are -

(A) Cl_2 is limiting reagent (B) 0.4 mole NaOH will left unreacted

(C) $[ClO_3^-]_{final\ solution} = 0.133$ M (D) $[Cl^-]_{final\ solution} = 0.267$ M **RR0214**

9. 100 ml of 0.4 M -acidified $KMnO_4$ solution may be decolourised completely by

(A) 200 ml 1N - $K_2Cr_2O_7$ solution (B) 300 ml 0.5M - H_2O_2 solution

(C) 100 ml 0.8N - KI solution (D) 75 ml 1.4 N - $H_2C_2O_4$ solution **RR0215**

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ASSERTION REASON TYPE :

10. **Statement–1 :-** Moles of $KMnO_4$ required for oxidation of Fe^{2+} in acidic and basic medium will be different.

Statement–2 :- Final oxidation state to which Mn^{7+} will be reduced will be different in case of acidic and basic medium.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true. **RR0131**

11. **Statement–1 :-** Mass of a particular substance that combine with 8 gm of oxygen is said to be equivalent weight of substance.

Statement–2 :-x gm of metal gave y gm of its oxide, so equivalent weight of metal is $\left(\frac{x}{y-x}\right)\times 8$

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true. **RR0132**

12. **Statement–1 :-** Degree of hardness of water is measured in terms of ppm of $CaCO_3$.

Statement–2 :- If water contains 120 ppm of $MgSO_4$, and 2ppm NaCl, its hardness in terms of $CaCO_3$ > 100 ppm.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true. **RR0133**

MATCH THE COLUMN

13.

Column-I	**Column-II**
(A) $P_2H_4 \longrightarrow PH_3 + P_4H_2$	(p) $E = \frac{3M}{4}$
(B) $I_2 \longrightarrow I^- + IO_3^-$	(q) $E = \frac{3M}{5}$
(C) $MnO_4^- + Mn^{2+} + H_2O \longrightarrow \underline{Mn_3O_4} + H^+$	(r) $E = \frac{15M}{26}$
(D) $\underline{H_3PO_2} \longrightarrow PH_3 + H_3PO_3$	(s) $E = \frac{5M}{6}$ **RR0134**

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Paragraph for Q.14 to Q.17

Equivalent weight $= \frac{\text{Molecular weight / Atomic weight}}{\text{n – factor}}$

n-factor is very important in redox as well as non-redox reactions. With the help of n-factor we can predicts the molar ratio of the reactant species taking part in reactions. The reaciprocal of n-factor's ratio of the reactants is the molar ratio of the reactants.

In general n-factor of acid/base is number of moles of H^+ /OH^- furnished per mole of acid/base. n-factor of a reactant is no. of moles of electrons lost or gained per mole of reactant.

Example 1 :

1. In acidic medium : $KMnO_4$ (n = 5) $\longrightarrow Mn^{2+}$
2. In neutral medium : $KMnO_4$ (n = 3) $\longrightarrow Mn^{2+}$
3. In basic medium : $KMnO_4$ (n = 1) $\longrightarrow Mn^{6+}$

Example 2 : $FeC_2O_4 \longrightarrow Fe^{3+} + 2CO_2$

Total no. of moles of e^- lost by 1 mole of $FeC_2O_4 = 1 + 1 \times 2 \Rightarrow 3$

$\therefore$ n-factor of $FeC_2O_4 = 3$

14. n-factor of $Ba(MnO_4)_2$ in acidic medium is :

(A) 2 (B) 6 (C) 10 (D) none of these

RR0135

15. For the reaction,

$$H_3PO_2 + NaOH \longrightarrow NaH_2PO_2 + H_2O$$

What is the equivalent weight of H_3PO_2 ? (mol. wt. is M)

(A) M (B) M/2 (C) M/3 (D) none of these

RR0136

16. For the reaction, $Fe_{0.95}O$ (molar mass : M) $\longrightarrow Fe_2O_3$. What is the eq. wt. of $Fe_{0.95}O$?

(A) $\frac{M}{0.85}$ (B) $\frac{M}{0.95}$ (C) $\frac{M}{0.8075}$ (D) none of these

RR0137

17. In the reaction, $xVO + yFe_2O_3 \longrightarrow FeO + V_2O_5$. What is the value of x and y respectively ?

(A) 1, 1 (B) 2, 3 (C) 3, 2 (D) none of these

RR0138

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TABLE TYPE QUESTION :

Column-I (1 mole of each oxidizing agent is taken)	**Column-II (oxidation number**	**Column-III (Reducing agents)**
(A) $KMnO_4$ (acidic)	(P) O.N of Mn = 4	(1) 3 mole of $FeSO_4$
(B) $KMnO_4$ (neutral)	(Q) O.N of Mn = 7	(2) 0.5 mole of I_2 changes to HIO_3
(C) MnO_2 (acidic)	(R) O.N of Cr = 6	(3) 1 mole of $K_2C_2O_4$
(D) K_2CrO_4 (acidic)	(S) O.N of Cr = 7	(4) 1.5 mole K_2SO_3

18. Which of the following is correct ?

(A) A ; P ; 2 (B) A ; Q ; 4 (C) B ; Q ; 1 (D) B ; Q ; 3 **RR0139**

19. Which of the following is correct ?

(A) A ; Q ; 3 (B) C ; P ; 3 (C) C ; P ; 4 (D) C ; Q ; 1 **RR0139**

20. Which of the following is/are correct ? (one or more than one correct)

(A) D ; R ; 1 (B) D ; R ; 2 (C) D ; S ; 1 (D) D ; R ; 4 **RR0139**

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EXERCISE # J-MAINS

1. Consider the reaction **[JEE(Main-online)-2014]**

$$H_2SO_{3(aq)} + Sn^{4+}_{(aq)} + H_2O_{(l)} \rightarrow Sn^{2+}_{(aq)} + HSO^-_{4(aq)} + 3H^+_{(aq)}$$

Which of the following statements is correct?

(1) H_2SO_3 is the reducing agent because it undergoes oxidation

(2) H_2SO_3 is the reducing agent because it undergoes reduction

(3) Sn^{4+} is the reducing agent because it undergoes oxidation

(4) Sn^{4+} is the oxidizing agent because it undergoes oxidation **RR0143**

2. In which of the following reaction H_2O_2 acts as a reducing agent ? **[JEE(Main)-2014]**

(a) $H_2O_2 + 2H^+ + 2e^- \rightarrow 2H_2O$ (b) $H_2O_2 - 2e^- \rightarrow O_2 + 2H^+$

(c) $H_2O_2 + 2e^- \rightarrow 2OH^-$ (d) $H_2O_2 + 2OH^- - 2e^- \rightarrow O_2 + 2H_2O$

(1) (a), (c) (2) (b), (d) (3) (a), (b) (4) (c), (d) **RR0144**

3. The molecular formula of a commercial resin used for exchanging ions in water softening is $C_8H_7SO_3Na$ (Mol. w.t 206). What would be the maximum uptake of Ca^{2+} ions by the resin when expressed in mole per gram resin ? **[JEE(Main)-2015]**

(1) $\frac{2}{309}$ (2) $\frac{1}{412}$ (3) $\frac{1}{103}$ (4) $\frac{1}{206}$ **RR0145**

4. The volume of 0.1N dibasic acid sufficient to neutralize 1 g of a base that furnishes 0.04 mole of OH^- in aqueous solution is : **[JEE(Main)-OnLine-2016]**

(1) 400 mL (2) 200 mL (3) 600 mL (4) 800 mL **RR0146**

5. Which of the following reactions is an example of a redox reaction ? **[JEE(Main)-2017]**

(1) $XeF_4 + O_2F_2 \rightarrow XeF_6 + O_2$ (2) $XeF_2 + PF_5 \rightarrow [XeF]^+PF_6^-$

(3) $XeF_6 + H_2O \rightarrow XeOF_4 + 2HF$ (4) $XeF_6 + 2H_2O \rightarrow XeO_2F_2 + 4HF$ **RR0147**

6. In KO_2, the nature of oxygen species and the oxidation state of oxygen atom are, respectively

(1) Superoxide and –1/2 (2) Oxide and -–2 **[JEE(Main)-OnLine-2018]**

(3) Peroxide and –1/2 (4) Superoxide and –1 **RR0150**

7. The hardness of a water sample (in terms of equivalents of $CaCO_3$) containing 10^{-3} M $CaSO_4$ is: (molar mass of $CaSO_4$ = 136 g mol^{-1}) **[JEE(Main)-(Jan.)-2019]**

(1) 100 ppm (2) 50 ppm (3) 10 ppm (4) 90 ppm **RR0153**

8. 50 mL of 0.5 M oxalic acid is needed to neutralize 25 mL of sodium hydroxide solution. The amount of NaOH in 50 mL of the given sodium hydroxide solution is : **[JEE(Main)-(Jan.)-2019]**

(1) 4 g (2) 2 g (3) 8 g (4) 1 g **RR0154**

9. In order to oxidise a mixture one mole of each of FeC_2O_4, $Fe_2(C_2O_4)_3$, $FeSO_4$ and $Fe_2(SO_4)_3$ in acidic medium , the number of moles of $KMnO_4$ required is - **[JEE(Main)-(April)-2019]**

(1) 3 (2) 2 (3) 1 (4) 1.5 **RR0155**

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10. An example of a disproportionation reaction is : **[JEE(Main)-(April)-2019]**

(1) $2KMnO_4 \rightarrow K_2MnO_4 + MnO_2 + O_2$ (2) $2MnO_4^- + 10I^- + 16H^+ \rightarrow 2Mn^{2+} + 5I_2 + 8H_2O$

(3) $2CuBr \rightarrow CuBr_2 + Cu$ (4) $2NaBr + Cl_2 \rightarrow 2NaCl + Br_2$ **RR0158**

11. 100 mL of a water sample contains 0.81 g of calcium bicarbonate and 0.73 of magnesium bicarbonate. The hardness of this water sample expressed in terms of equivalents of $CaCO_3$ is:

(molar mass of calcium bicarbonate is 162 g mol^{-1} and magnesium bicarbonate is 146 $gmol^{-1}$)

(1) 1,000 ppm (2) 10,000 ppm **[JEE(Main)-(April)-2019]**

(3) 100 ppm (4) 5,000 ppm **RR0159**

12. 25 ml of the given HCl solution requires 30 mL of 0.1 M sodium carbonate solution. What is the volume of this HCl solution required to titrate 30 mL of 0.2 M aqueous NaOH solution?

(1) 25 mL (2) 50 mL **[JEE(Main)-(Jan)-2019]**

(3) 12.5 mL (4) 75 mL **RR0160**

13. Oxidation number of potassium in K_2O, K_2O_2 and KO_2 , respectively, is : **[JEE(Main)-(Jan)-2020]**

(1) +1, +4 and +2 (2) +1, +2 and +4 (3) +1, +1 and +1 (4) +2 , +1 and $+\frac{1}{2}$

RR0167

14. The compound that cannot act both as oxidising and reducing agent is :**[JEE(Main)-(Jan)-2020]**

(1) H_2O_2 (2) H_2SO_3 (3) HNO_2 (4) H_3PO_4

RR0168

15. The strength of an aqueous NaOH solution is most accurately determined by titrating :

(Note : consider that an appropriate indicator is used) **[JEE(Main)-(Jan)-2020]**

(1) Aq. NaOH in a volumetric flask and concentrated H_2SO_4 in a conical flask

(2) Aq. NaOH in a pipette and aqueous oxalic acid in a burette

(3) Aq. NaOH in a burette and concentrated H_2SO_4 in a conical flask

(4) Aq. NaOH in a burette and aqueous oxalic acid in a conical flask **RR0217**

16. A 20.0 mL solution containing 0.2 g impure H_2O_2 reacts completely with 0.316 g of $KMnO_4$ in acid solution. The purity of H_2O_2 (in %) is _____ (mol. wt. of H_2O_2 = 34; mol. wt. of $KMnO_4$ = 158) **[JEE(Main)-(Jan)-2020]**

RR0218

17. A 100 mL solution was made by adding 1.43 g of $Na_2CO_3 \cdot xH_2O$. The normality of the solution is 0.1 N. The value of x is _____. **[JEE(Main)-(Jan)-2020]**

(The atomic mass of Na is 23g/mol) :- **RR0219**

18. The volume, in mL, of 0.02 M $K_2Cr_2O_7$ solution required to react with 0.288 g of ferrous oxalate in acidic medium is_____. **[JEE(Main)-(Jan)-2020]**

(Molar mass of Fe = 56 g mol^{-1}) **RR0220**

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19. The reaction of sulphur in alkaline medium is the below: **[JEE(Main)-(Jan)-2021]**

$$S_{8(s)} + a\ OH^-_{(aq)} \rightarrow b\ S^{2-}_{(aq)} + c\ S_2O_3^{2-}{}_{(aq)} + d\ H_2O_{(\ell)}$$

The values of 'a' is _____. (Integer answer) **RR0222**

20. In basic medium CrO_4^{2-} oxidises $S_2O_3^{2-}$ to form SO_4^{2-} and itself changes into $Cr(OH)_4^-$. The volume of 0.154 M CrO_4^{2-} required to react with 40 mL of 0.25 M $S_2O_3^{2-}$ is _____ mL. (Rounded-off to the nearest integer) **[JEE(Main)-(Jan)-2021]**

RR0223

21. 10.0 ml of Na_2CO_3 solution is titrated against 0.2 M HCl solution. The following titre values were obtained in 5 readings.

4.8 ml, 4.9 ml, 5.0 ml, 5.0 ml and 5.0 ml **[JEE(Main)-(Jan)-2021]**

Based on these readings, and convention of titrimetric estimation of concentration of Na_2CO_3 solution is _____mM. (Round off to the Nearest integer) **RR0224**

22. 250 mL of 0.5 M NaOH was added to 500 mL of 1 M HCl. The number of unreacted HCl molecules in the solution after complete reaction is _____ $\times 10^{21}$. (Nearest integer)

($N_A = 6.022 \times 10^{23}$) **[JEE(Main)-(Jan)-2021]**

RR0225

23. When 10 mL of an aqueous solution of Fe^{2+} ions was titrated in the presence of dil H_2SO_4 using diphenylamine indicator, 15 mL of 0.02 M solution of $K_2Cr_2O_7$ was required to get the end point. The molarity of the solution containing Fe^{2+} ions is $x \times 10^{-2}$ M. The value of x is _____. (Nearest integer) **[JEE(Main)-(Jan)-2021]**

RR0226

24. When 10 mL of an aqueous solution of $KMnO_4$ was titrated in acidic medium, equal volume of 0.1 M of an aqueous solution of ferrous sulphate was required for complete discharge of colour. The strength of $KMnO_4$ in grams per litre is _____ $\times 10^{-2}$. (Nearest integer)

[Atomic mass of K = 39, Mn = 55, O = 16] **[JEE(Main)-(Jan)-2021]**

RR0227

25. 0.01 M $KMnO_4$ solution was added to 20.0 mL of 0.05 M Mohr's salt solution through a burette. The initial reading of 50 mL burette is zero. The volume of $KMnO_4$ solution left in the burette after the end point is _____ mL. (nearest integer) **[JEE(Main)-(June)-2022]**

RR0246

26. A 2.0 g sample containing MnO_2 is treated with HCl liberating Cl_2. The Cl_2 gas is passed into a solution of KI and 60.0 mL of 0.1 M $Na_2S_2O_3$ is required to titrate the liberated iodine. The percentage of MnO_2 in the sample is _____. (Nearest integer) [Atomic masses (in u) Mn = 55; Cl = 35.5; O = 16, I = 127, Na = 23, K = 39, S = 32] **[JEE(Main)-(June)-2022]**

RR0247

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27. In neutral or faintly alkaline medium, $KMnO_4$ being a powerful oxidant can oxidize, thiosulphate almost quantitatively, to sulphate. In this reaction overall change in oxidation state of manganese will be :

[JEE(Main)-(July)-2022]

(A) 5 (B) 1 (C) 0 (D) 3

RR0248

28. 20 mL of 0.02 M hypo solution is used for the titration of 10 mL of copper sulphate solution, in the presence of excess of KI using starch as an indicator. The molarity of Cu^{2+} is found to be _____ $\times 10^{-2}$ M [nearest integer]

Given : $2Cu^{2+} + 4I^- \longrightarrow Cu_2I_2 + I_2$

$I_2 + 2S_2O_3^{2-} \longrightarrow 2I^- + S_4O_6^{2-}$ **[JEE(Main)-(July)-2022]**

RR0249

29. In the titration of $KMnO_4$ and oxalic acid in acidic medium, the change in oxidation number of carbon at the end point is_____ **[JEE(Main)-(July)-2022]**

RR0250

30. 20 mL of 0.02 M $K_2Cr_2O_7$ solution is used for the titration of 10 mL of Fe^{2+} solution in the acidic medium. The molarity of Fe^{2+} solution is _____ $\times 10^{-2}$ M. (Nearest Integer) **[JEE(Main)-(July)-2022]**

RR0251

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EXERCISE # J-ADVANCED

1. Reduction of the metal centre in aqueous permanganate ion involves - **[JEE-2011]**

(A) 3 electrons in neutral medium (B) 5 electrons in neutral medium

(C) 3 electrons in alkaline medium (D) 5 electrons in acidic medium **RR0161**

2. Reaction of Br_2 with Na_2CO_3 in aqueous solution gives sodium bromide and sodium bromate with evolution of CO_2 gas. The number of sodium bromide molecules involved in the balanced chemical equation is. **[JEE- 2011]**

RR0162

3. Which ordering of compounds is according to the decreasing order of the oxidation state of nitrogen-

(A) HNO_3, NO, NH_4Cl, N_2 (B) HNO_3, NO, N_2, NH_4Cl

(C) HNO_3, NH_4Cl, NO, N_2 (D) NO, HNO_3, NH_4Cl, N_2 **[JEE- 2012]**

RR0163

4. 25 mL of household bleach solution was mixed with 30 mL of 0.50 M KI and 10 mL of 4 N acetic acid. In the titration of the liberated iodine, 48 mL of 0.25 N $Na_2S_2O_3$ was used to reach the end point. The molarity of the household bleach solution is **[JEE- 2012]**

(A) 0.48 M (B) 0.96 M (C) 0.24 M (D) 0.024 M **RR0164**

5. For the reaction **[JEE- 2014]**

$I^- + ClO_3^- + H_2SO_4 \rightarrow Cl^- + HSO_4^- + I_2$

The correct statement(s) in the balanced equation is / are :

(A) Stoichiometric coefficient of HSO_4^- is 6

(B) Iodide is oxidized

(C) Sulphur is reduced

(D) H_2O is one of the products **RR0228**

6. In neutral or faintly alkaline solution, 8 moles permanganate anion quantitatively oxidize thiosulphate anions to produce X moles of a sulphur containing product. the magnitude of X is **[JEE- 2016]**

RR0165

7. The order of the oxidation state of the phosphorus atom in H_3PO_2, H_3PO_4, H_3PO_3 and $H_4P_2O_6$ is **[JEE- 2017]**

(A) $H_3PO_4 > H_4P_2O_6 > H_3PO_3 > H_3PO_2$ (B) $H_3PO_3 > H_3PO_2 > H_3PO_4 > H_4P_2O_6$

(C) $H_3PO_2 > H_3PO_3 > H_4P_2O_6 > H_3PO_4$ (D) $H_3PO_4 > H_3PO_2 > H_3PO_3 > H_4P_2O_6$

RR0229

8. To measure the quantity of $MnCl_2$ dissolved in an aqueous solution, it was completely converted to $KMnO_4$ using the reaction, **[JEE- 2018]**

$MnCl_2 + K_2S_2O_8 + H_2O \rightarrow KMnO_4 + H_2SO_4 + HCl$ (equation not balanced).

Few drops of concentrated HCl were added to this solution and gently warmed. Further, oxalic acid (225 mg) was added in portions till the colour of the permanganate ion disappeard. The quantity of $MnCl_2$ (in mg) present in the initial solution is _____.

(Atomic weights in g mol^{-1} : Mn = 55, Cl = 35.5) **RR0166**

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9. The amount of water produced (in g) in the oxidation of 1 mole of rhombic sulphur by conc.HNO_3 to a compound with the highest oxidation state of sulphur is ___

(Given data : Molar mass of water = 18 g mol^{-1}) **RR0230**

10. 5.00 mL of 0.10 M oxalic acid solution taken in a conical flask is titrated against NaOH from a burette using phenolphthalein indicator. The volume of NaOH required for the appearance of permanent faint pink color is tabulated below for five experiments. What is the concentration, in molarity, of the NaOH solution ? **[JEE- 2020]**

Exp. No.	Vol. of NaOH (mL)
1	12.5
2	10.5
3	9
4	9
5	9

RR0231

11. In the chemical reaction between stoichiometric quantities of $KMnO_4$ and KI in weakly basic solution, what is the number of moles of I_2 released for 4 moles of $KMnO_4$ consumed ? **[JEE- 2020]**

RR0232

Question Stem for Question Nos. 12 and 13

Question Stem

A sample (5.6 g) containing iron is completely dissolved in cold dilute HCl to prepare a 250 mL of solution. Titration of 25.0 mL of this solution requires 12.5 mL of 0.03 M $KMnO_4$ solution to reach the end point. Number of moles of Fe^{2+} present in 250 mL solution is $x \times 10^{-2}$ (consider complete dissolution of $FeCl_2$). The amount of iron present in the sample of y% by weight.

(Assume : $KMnO_4$ reacts only with Fe^{2+} in the solution **[JEE- 2021]**

Use : Molar mass of iron as 56 g mol^{-1})

12. The value of x is _____. **RR0233**

13. The value of y is _____. **RR0234**

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ANSWER KEY

EXERCISE # S-I

1. (a) +5 (b) +2 (c) –1 (d) +2
(e) +6 (f) +3 (g) +4/3 (h) –1/3
(i) +8/3 (j) +16/3 (k) 0 (l) +1/2
(m) –1/2 (n) +2

2. (a) +3 (b) +3, +6 (c) +2 (d) $+\frac{200}{93}$
(e) +6 (f) +6 (g) +6

3. (a) Reduction (b) Oxidation (c) Reduction (d) Oxidation
(e) Reduction (f) Oxidation (g) Reduction

4. (a) Oxidant : O_2 ; Reductant : Zn
(b) Oxidant : H^+ ; Reductant : Zn
(c) Oxidant : NO_3^- ; Reductant : Zn
(d) Oxidant : SO_2 ; Reductant : H_2S
(e) Oxidant : MnO_4^- ; Reductant : Cl^-
(f) Oxidant : MnO_4^- ; Reductant : H_2O_2

5. (i) $Cu + 4HNO_3 \longrightarrow Cu(NO_3)_2 + 2NO_2 + 2H_2O$
(ii) $Br_2 + 5H_2O_2 \longrightarrow 2BrO_3^- + 4H_2O + 2H^+$
(iii) $MnO_4^- + 5Fe^{+2} + 8H^+ \longrightarrow Mn^{+2} + 5Fe^{+3} + 4H_2O$
(iv) $MnO_2 + 2Cl^- + 4H^+ \longrightarrow Mn^{+2} + Cl_2 + 2H_2O$
(v) $3ClO^- + 2CrO_2^- + 2OH^- \longrightarrow 3Cl^- + 2CrO_4^{2-} + H_2O$
(vi) $Zn + 2NaOH \longrightarrow Na_2ZnO_2 + H_2$
(vii) $2Na_2S_2O_3 + I_2 \longrightarrow Na_2S_4O_6 + 2NaI$

6. (i) $2MnO_4^- + 6I^- + 4H_2O \longrightarrow 2MnO_2 + 3I_2 + 8OH^-$
(ii) $H^+ + 2MnO_4^- + 5SO_2 + 2H_2O \longrightarrow 2Mn^{2+} + 5HSO_4^-$
(iii) $2Fe^{2+} + H_2O_2 + 2H^+ \longrightarrow 2Fe^{3+} + 2H_2O$
(iv) $Cr_2O_7^{2-} + 3SO_2 + 2H^+ \longrightarrow 2Cr^{3+} + 3SO_4^{2-} + H_2O$
(v) $P_4 + 3OH^- + 3H_2O \longrightarrow PH_3 + 3H_2PO_2^-$
(vi) $3MnO_4^- + 5FeC_2O_4 + 24H^+ \longrightarrow 3Mn^{2+} + 5Fe^{3+} + 10CO_2 + 12\ H_2O$
(vii) $2S_2O_3^{2-} + I_2 \longrightarrow S_4O_6^{2+} + 2I^-$

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7. (1) 1 (2) 1 (3) 1 (4) 1 (5) 2

8. (i) 98, (ii) 49, (iii) 32.66 or 32.67

9. Ans.21.00

10. Ans. (i) 13.91 or 13.92 (ii) 5.66 or 5.67

11. Ans.5

12. (i) 71.00 (ii) 143.00 (iii) 44.50 (iv) 57.00
(v) 93.66 or 93.67

13. (i) 0.79 (ii) $\frac{1.58}{0.93}$ = 1.69 or 1.70 (iii) 0.66 or 0.67 (iv) 1.00

14. (i) 3 (ii) 11 (iii) 27 (iv) 16

15. (i) 98.00 (ii) 98.00 (iii) 98.00 (iv) 98.00

16. 80.00 **17.** 40.00 **18.** 40.00 **19.** 12.50

20. 3.00 **21.** 200.00 **22.** 35.08 or 35.09 **23.** 500.00

24. 0.05 **25.** 16.00 **26.** 4.54 **27.** 10.00

28. 2.12 **29.** 0 **30.** 160.00 **31.** 0.58 or 0.59

32. 4.00 **33.** 16.66 or 16.67 **34.** 3.00 **35.** 200.00

36. 25.00 **37.** 2.50 **38.** 0.42 **39.** 7.00

40. 30.00 **41.** 20.00 **42.** 11.20 **43.** 4.00

44. 100.00 **45.** 250.00

EXERCISE # S-II

1. (a) –3, –1 (b) –3, +3 (c) 0 (d) +1
(e) +3 (f) –3 (g) +3 (h) + 2
(i) +2 (j) +4

2. (i) $6KMnO_4(aq.) + 10FeC_2O_4(aq.) + 24H_2SO_4(aq.) \longrightarrow$
$6MnSO_4(aq.) + 5Fe_2(SO_4)_3(aq.) + 20CO_2(g) + 3K_2SO_4 + 24H_2O(l)$

(ii) $3HNO_2(aq.) \rightarrow NO_3^- + H^+ + 2NO(g) + H_2O$

(iii) $IO_3^- + 5I^- + 6H^+ \rightarrow 3I_2 + 3H_2O$

(iv) $5Cu_2S + 8MnO_4^- + 44H^+ \rightarrow 10Cu^{2+} + 5SO_2 + 8Mn^{2+} + 22\ H_2O$

(v) $HCHO(l) + 2[Ag(NH_3)_2]^+(aq.) + 3OH^-(aq.) \rightarrow 2Ag(s) + 4NH_3(aq.) + HCOO^-(aq.) + 2H_2O(l)$

3. 6 **4.** 40.00 **5.** 2.50 **6.** 3.00

7. 1.00 **8.** 39.42 **9.** 5.00 **10.** 250

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EXERCISE # O-I

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	C	B	B	D	A	D	A	A	D	A
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	A	B	C	C	A	C	C	A	A	D
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	D	B	C	D	A	C	C	D	D	B
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	A	C	A	C	A	B	C	C	B	D

EXERCISE # O-II

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A	B	B	C	D	A,B,D	A,B,C	B,D	B	A
Que.	11	12	13				14	15	16	17
Ans.	B	C	A-s	B-q	C-r	D-P	C	A	A	B
Que.	18	19	20							
Ans.	C	B	A,D							

EXERCISE # J-MAINS

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	1	2	2	1	1	1	1	1	2	3
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	2	1	3	4	4	85	10	50.00	12	173
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	50	226	18	316	30	13	D	4	1	24

EXERCISE # J-ADVANCED

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A,C,D	5	B	C	A,B,D	6	A	126	288.00	0.11
Que.	11	12	13							
Ans.	6	1.87 or 1.88	18.75							

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Important Notes

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